

Lab 17 - Controllers & Actions

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Assignment: Lab 17 | ITI Summer Training - Web Development Using .NET

Fingerprinted values for Lab ID 31

Variable Name	Formula / Derivation	Result Value
MIN_GPA_EDIT	$2.0 + (\text{Lab ID mod } 5) * 0.3 = 2.0 + 1 * 0.3$	2.3
MAX_YEAR_EDIT	$(\text{Lab ID mod } 3) + 2 = 1 + 2$	3

Part A - Verify and Orient

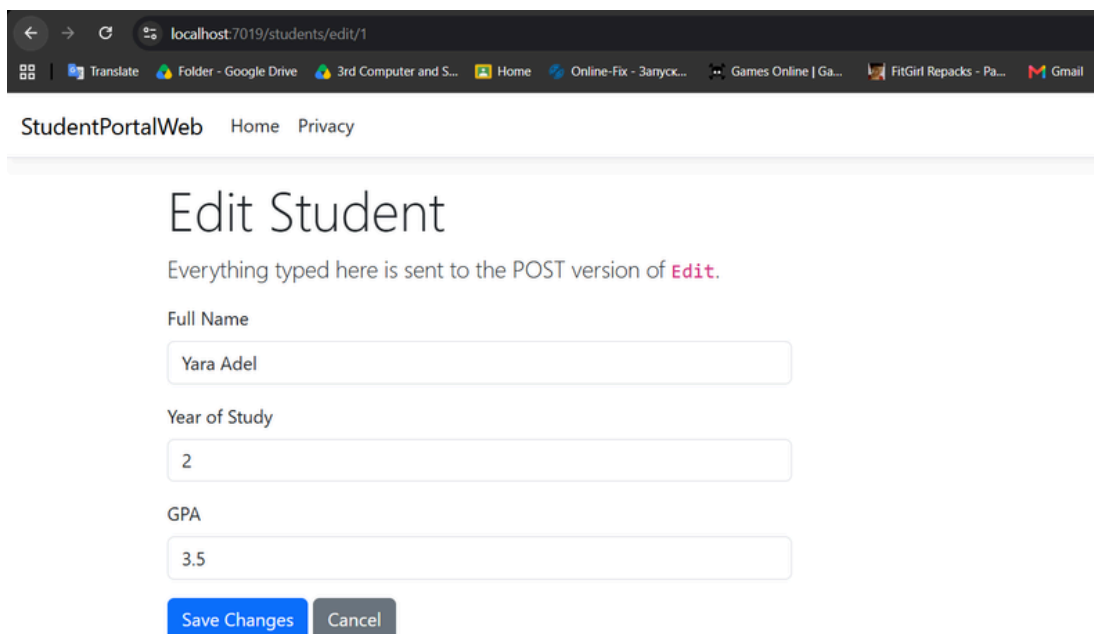
Verified that conventional and attribute routes behave correctly. Added fingerprinted values header comment to StudentsController.cs.

- **Why is Create()'s GET overload not marked async, even though the POST overload is?** The GET overload doesn't perform any database queries or asynchronous file operations (it just returns the view), so there is no task to await. The POST overload runs AddAsync and SaveChangesAsync to update the database, requiring the async/await keyword pattern.

Part B - The GET half: show the real row

Implemented Edit (int id) GET action to fetch the student from the database and pass it to pre-populate Edit.cshtml. Returns 404 if the record is missing.

- **Why does this action load the student from the database?** It loads the student from the database to pre-populate the input fields with their current values, whereas Create() is for a new record and starts with empty inputs.



← → ↻ 📑 localhost:7019/students/edit/1

🔍 Translate 📁 Folder - Google Drive 🖨️ 3rd Computer and S... 🏠 Home 🌐 Online-Fix - 3anyok... 🎮 Games Online | Ga... 📁 FitGirl Repacks - Pa... 📧 Gmail

StudentPortalWeb Home Privacy

Edit Student

Everything typed here is sent to the POST version of **Edit**.

Full Name

Year of Study

GPA

Save Changes Cancel

Part C - The POST half: save the change, properly

Implemented Edit (int id, Student student) POST action. It validates match boundaries, updates properties on the fetched DbContext entity, calls SaveChangesAsync, sets confirmation TempData, and redirects.

- **What would happen if you renamed its id parameter to studentId and left the route as {id}?** The model binder matches parameters by name. Since it cannot find a route parameter named 'studentId', it would silently assign the default value of 0, triggering the security/guard mismatch checks.

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Nada Samir was updated successfully.

All Students

Reached through the route pattern `students`.

+ Add a Student

Id	Full Name	Year	GPA	Actions
4	Kareem Fouad	4	3.20	Edit
3	Nada Samir	2	4.00	Edit
2	Omar Hesham	3	2.80	Edit
1	Yara Adel	2	3.50	Edit

Part D - Your own validation boundaries

Updated validation attributes on Student: YearOfStudy range 1-3, GPA range 2.3-4.0. Validated that C# validates this before the DB gets called.

- **Is this a schema attribute or validation-only, and how would you prove it?** Validation-only. You can prove it because modifying [Range] does not cause Entity Framework to detect changes or require migrations, whereas [Required] or [MaxLength] alterations generate column changes.

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Edit Student

Everything typed here is sent to the POST version of `Edit`.

Full Name

Year of Study

GPA

GPA must be between 2.30 and 4.00.

[Save Changes](#) [Cancel](#)

133 % No issues found

Results Messages

	Id	FullName	YearOfStudy	Gpa
1	1	Yara Adel	2	3.5
2	2	Omar Hesham	3	2.8
3	3	Nada Samir	2	4
4	4	Kareem Fouad	4	3.2

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Omar Hesham was updated successfully.

All Students

Reached through the route pattern `students`.

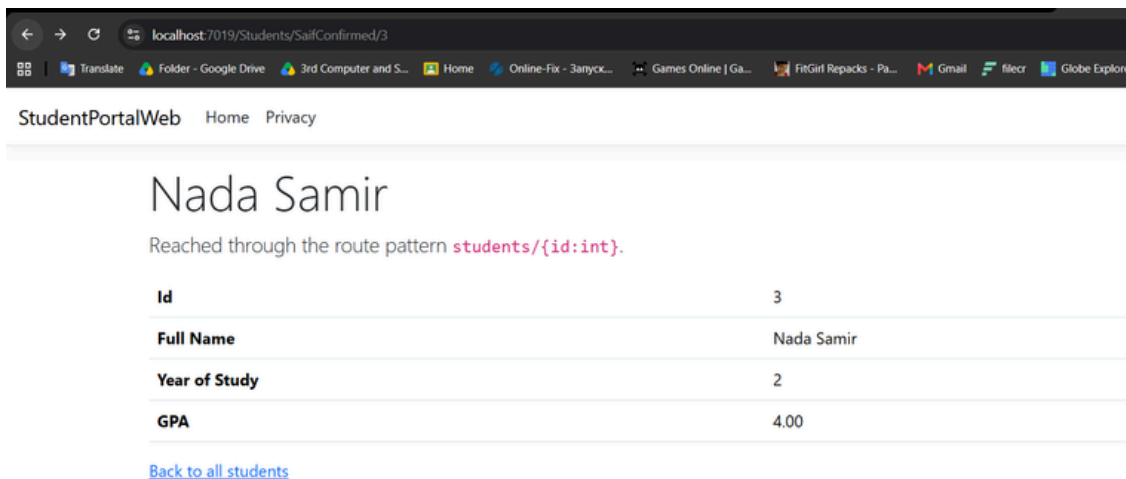
+ Add a Student

Id	Full Name	Year	GPA	Actions
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2	Omar Hesham	3	2.30	Edit
1	Yara Adel	2	3.50	Edit

Part E - One action that carries your own name

Added GET confirmation action **SaifConfirmed(int id)** that reloads the student from the database and renders the Details view. Successful Edit POST redirects to this action.

- **Why must this action reload the student from the database?** Reloading from the database guarantees that the details page shows the actual committed state of the data in the database, verifying the database write was fully successful.



Part F - Wrap-Up Reflection

1. **DerivedValues:** Lab ID: 31 | MIN_GPA_EDIT: 2.3 | MAX_YEAR_EDIT: 3.

2. **Three Checkpoints in Order:**

- Checkpoint 1: Route Constraints (URL checking before entering action)
- Checkpoint 2: Model Validation / ModelState (C# checking before action code starts)
- Checkpoint 3: Database Constraints (SQL Server integrity checks at save)
- Our Part D validation attributes live in checkpoint 2 (Model Validation).

3. **Browser Redirect Mechanism:**

- Under the RedirectToAction flow, the server returns a 302 Redirect telling the browser to navigate to /Students/SaifConfirmed/3.
- The browser updates the address bar and requests that URL via a safe, read-only GET request.
- When F5 is pressed, the browser simply re-triggers the GET request. There is no POST payload or save command, which prevents duplicate database inserts.

4. **Required/MaxLength vs. Range:**

- **Same:** Both are defined as attributes in C# above property fields.
- **Different:** Required/MaxLength are schema attributes changing database structure via migrations. Range is a validation-only attribute checked purely by C# at runtime without altering the database schema.